

External Modulation of an RF Signal Enhanced by Mutual Optical Injection Locking

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Abstract We demonstrate a novel external modulation scheme that increases the modulation efficiency for RF signals. It uses an EAM placed between two mutually injection-locked DFB lasers. Driving the EAM at the relaxation frequency of DFB lasers provides a modulation gain of 14 dB. ©2026 The Authors

Introduction

Microwave Photonics (MWP) is an interdisciplinary field that bridges radio frequency (RF) technology and photonics [1]. For MWP to be attractive, the size, weight, power consumption, and cost of its devices must be at least comparable to those of Monolithic Microwave Integrated Circuits (MMICs). Integrating MWP into Photonic Integrated Circuits (PICs) has enabled significant reductions in size and weight. However, MWP still lags behind MMICs when it comes to power consumption.

The power consumption of an integrated MWP circuit is determined by its constituent stages, starting with the electrical-to-optical conversion stage [2]. The simplest method for converting RF signals to light is through direct modulation. However, this approach requires large driving currents, and high modulation indices can induce nonlinear effects that degrade the spectral purity of the generated RF signals [3].

External modulation enables overcoming the limitations of direct modulation. Phase or Mach-Zehnder modulators, based on Pockels effect, enable efficient modulation, however at the cost of long device lengths so that a low voltage swing (V_π) is obtained. Electro-absorption modulators (EAMs), which operate via the Franz-Keldysh or Quantum-Confined Stark effect, offer a more compact, high-bandwidth alternative [4]. Due to its small size and compatible fabrication processes, an EAM can be integrated with a distributed feedback (DFB) laser, forming a structure known as an electro-absorption-modulated laser (EML) [5].

The modulation efficiency of optical modulators can be enhanced at the expense of reduced bandwidth, by exploiting resonance in either the electrical or optical domain. In the electrical domain, [6] demonstrates an RF driver amplifier that is impedance-matched to an EAM within a limited electric bandwidth. In the optical domain, [7, 8] report the use of ring resonators combined with external modulators to increase modulation effi-

ciency at specific optical frequencies. Although both approaches are complementary in exploring either electrical or optical resonances, neither includes the laser source, and therefore does not leverage its optical gain.

This paper proposes a novel external modulation scheme suitable for RF signals, which leverages the relaxation frequency of a pair of mutually injection-locked lasers. While being complementary with existing resonance-based approaches such as RF power amplifiers [6] and ring resonators [8], the proposed method increases modulation efficiency, reduces noise, and improves wall-plug efficiency. The proposed modulation scheme is compatible with EML fabrication processes, paving the way for efficient integrated MWP devices.

Operation principle

Figure 1 illustrates the operation principle. The system consists of two DFB lasers connected by an EAM. First, the two DFB lasers are tuned to the same emitting wavelength, so that mutual optical injection locking (MOIL) is obtained. Subsequently, the EAM is modulated by an RF signal. The relaxation frequency of each laser is then tuned to match the RF frequency. As a result, the RF sidebands of the modulated signal are amplified.

The main advantage of this approach is that op-

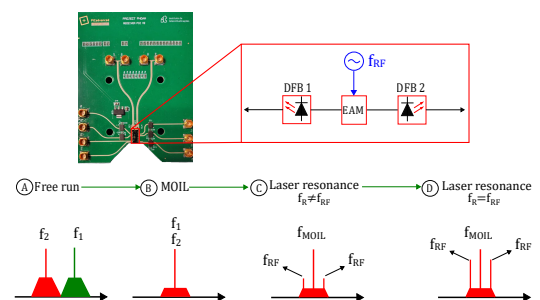


Fig. 1: Top. Assembled PIC and block diagram of the proposed external modulation scheme for RF signals. Bottom: operation principle. A. DFBs start in free-run. B. DFBs become locked. C. Laser resonance frequency is not yet adjusted to the RF frequency. D. Laser resonance frequency is adjusted to the RF frequency.

tical amplification, provided by the DFB lasers, is directed to the sidebands rather than to the optical carrier. Once the DFB lasers are locked, their linewidth is reduced, which therefore decreases phase noise. The outputs from the two DFB lasers can be combined, potentially doubling the overall optical power. The proposed modulation scheme can operate at different RF frequencies, and even at multiple RF frequencies (i.e., multi-band operation), as the DFB lasers can be independently biased to different relaxation frequencies [9]. Finally, the proposed modulation scheme is similar to widely deployed EMLs, which means that its fabrication process is already mature.

Comparison with other modulation schemes

Table 1 compares the proposed modulation scheme with other modulation schemes.

When one resorts to a semiconductor optical amplifier (SOA) for boosting a modulated RF signal, most of the added power is allocated to the optical carrier rather than to the RF sidebands. Consequently, a SOA-based modulation scheme is inherently inefficient. A SOA also adds amplification noise and therefore reduces signal-to-noise ratio.

The proposed scheme combines the intrinsic gain of the active media of two DFB lasers with an additional contribution from resonance, resulting in an RF sideband amplification gain of G_{RF} . Consequently, the wall-plug efficiency increases by the same factor.

In terms of noise, the proposed scheme achieves noise reduction, owing to MOIL.

Numerical validation

In this section, the operation principle behind this concept is numerically demonstrated. The objective is to estimate the relative gain provided to the RF sidebands, G_{RF} .

The proposed modulation scheme can be modelled by the following coupled differential delay equations [3, 10]:

$$\frac{dE_1}{dt} = \frac{1}{2}(1 + i\alpha)g(N_1 - N_{th})E_1 + \frac{\kappa}{\tau_{in}}s_{\text{EAM}}(t)E_2(t - \tau) \exp(i\omega_2\tau + i\Delta\omega t), \quad (1)$$

$$\frac{dN_1}{dt} = \frac{I_1}{eV} - \tau_c^{-1}N_1 - \Gamma g(N_1 - N_0)|E_1|^2, \quad (2)$$

$$\frac{dE_2}{dt} = \frac{1}{2}(1 + i\alpha)g(N_2 - N_{th})E_2 + \frac{\kappa}{\tau_{in}}s_{\text{EAM}}(t)E_1(t - \tau) \exp(i\omega_1\tau - i\Delta\omega t), \quad (3)$$

$$\frac{dN_2}{dt} = \frac{I_2}{eV} - \tau_c^{-1}N_2 - \Gamma g(N_2 - N_0)|E_2|^2, \quad (4)$$

where E_i represents the electric field of each laser, I_i is the bias current, N_i is the corresponding carrier density, and ω_i is the angular frequency, for $i = 1, 2$. α is assumed to be 3, in line with [11].

Tab. 1: Comparison between modulation schemes based on: relative RF sideband gain, G , relative wall-plug efficiency, η , and noise, N .

Mod. Scheme	G	η	Noise
DFB + mod	0	0	N_{DFB}
DFB + mod + SOA	G_{SOA}	$\propto G_{\text{SOA}}$	$\gg N_{\text{DFB}}$
DFB + mod + DFB	G_{RF}	$\propto G_{\text{RF}}$	$< N_{\text{DFB}}$

This is a conservative estimation, as it reduces gain when compared to higher values. As for the remaining parameters, given that they are not known by the authors, parameter values listed in [10] were thus considered here. For simplicity, the modulation function of the EAM is approximated to first order, such that $s_{\text{EAM}}(t) = A_0 + A_{\text{RF}} \sin(\omega_{\text{RF}}t)$, where ω_{RF} is the RF frequency. A_0 and A_{RF} represent the amplitudes of the optical carrier and the RF driving signal, respectively, and together they determine the modulation depth. Finally, κ is the coupling strength (i.e., injection ratio), $\Delta\omega$ is the frequency detuning between lasers, and τ is the delay between lasers.

The relaxation frequency of a free-run laser is proportional to $\sqrt{I}$. The bias currents of both lasers were tuned to achieve free-running relaxation frequencies of $f_{R1} = 6$ GHz and $f_{R2} = 8.5$ GHz. Such relaxation frequencies were chosen to match the experimental values.

The first step to determine the relative sideband gain, G_{RF} , is to obtain a reference sideband power, in turn estimated from the output signal of a DFB+mod modulation scheme. To do this, the rate equations were first numerically solved under free-running conditions, i.e., $\kappa = 0$. Modulation was then introduced by setting $A_0 = 1$ and $A_{\text{RF}} = 0.01$. The modulator function $s_{\text{EAM}}(t)$ was multiplied with the obtained optical field $E(t)$ to simulate the modulated output.

Next, the rate equations were solved for the coupled system under modulation, i.e., $\kappa \neq 0$. The coupling strength κ and frequency detuning $\Delta\omega$ were independently swept. Initially, both lasers were set to the same bias points, resulting in identical relaxation frequencies. The EAM was driven at this common frequency, i.e., $f_{\text{RF}} = f_{R1} = f_{R2} = 6$ GHz. Then, to closely reflect realistic conditions [10], the lasers were subsequently biased at different currents, resulting in distinct relaxation frequencies at $f_{R1} = 6$ GHz and $f_{R2} = 8.5$ GHz. In this scenario, the EAM was modulated at $f_{\text{RF}} = f_{R1} = 6$ GHz. For a first iteration, the delay between lasers was set to zero, $\tau = 0$.

Figure 2 presents the numerical results, showing that the locking range is symmetric when the relaxation frequencies are equal, but exists only for positive detunings when the relaxation frequen-

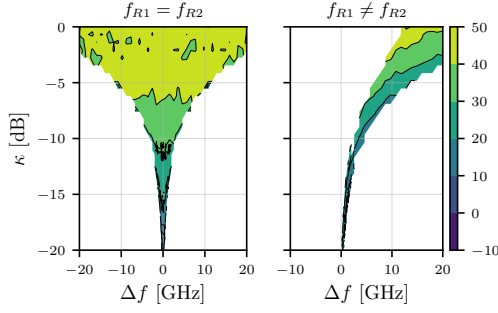


Fig. 2: Sideband gain for an EAM driven at $f_{RF} = f_{R2} = 6$ GHz as a function of κ and $\Delta\omega$, shown for both $f_{R1} = f_{R2}$ and $f_{R1} \neq f_{R2}$ cases.

cies differ. In both cases, the relative sideband gain increases with the injection ratio κ , reaching an upper bound of $G_{RF,max} = 48$ dB.

In a second iteration, the impact of the delay between the lasers was considered. The delay reduces the achievable locking range from top to bottom. This means that injection locking is no longer possible for high injection ratios, which therefore reduces the maximum achievable gain. Further analysis is required to fully assess the effect of this delay.

In the experimental setup, the delay is $\tau = 100$ ps since DFB1, the EAM, and DFB2 do not belong to a single active segment. Instead, they are connected by passive waveguides, and culminating in an injection of only -18 dB. Under these conditions, the corresponding frequency response was evaluated. The results are presented in Figure 4. The numerically estimated gains at f_{R1} and f_{R2} are 19 dB and 16 dB, respectively.

Experimental Validation

The PIC used to demonstrate the proposed modulation scheme is shown in Figure 1. It was fabricated using InP monolithic integration standard building blocks available from a multi-project wafer run offered by Fraunhofer HHI.

Before driving the EAM, the DFBs were first mutually locked, as confirmed in Figure 3. As predicted, noise near the carrier is reduced compared with free-running operation, and new side modes appear. These side modes correspond to the relaxation oscillation frequencies of each laser and the associated nonlinear processes determine the appearance of the side modes.

The frequency response of the proposed modulation scheme was then measured by modulating the EAM with an RF tone at 0 dBm with variable RF frequency, and recording the optical spectra. Although not shown in Figure 1, DFB2 also serves as a laser source for three other EAMs. Consequently, the frequency response of DFB2 in free-run operation (with DFB1 turned off) with each of these three EAMs was measured as well.

Figure 4 presents the results. Local maxima

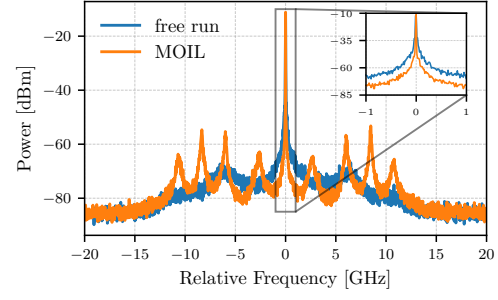


Fig. 3: Comparison between free-run lasing and MOIL. The optical spectra were obtained with a resolution of 13 MHz.

are observed at 2.5 GHz, 6 GHz, 8.5 GHz, and 10.5 GHz. The peaks at 6 GHz and 8.5 GHz correspond to the relaxation frequencies of DFB1 and DFB2, with respective gains of 14 dB and 9 dB. The experimental gains are lower than the numerical predictions of 19 dB and 16 dB, primarily due to discrepancies between the parameters used in the simulation and the actual device parameters. The numerical results do not predict resonances at 2.5 GHz and 10.5 GHz. This is because the numerical model (1)–(4) does not capture nonlinear phenomena or parasitic reflections at the ends of the DFBs.

Conclusion

This paper presents a novel modulation scheme suitable for RF signals using two mutually injection-locked DFB lasers connected by an EAM. Significant gain can be obtained at improved wall-plug efficiency and with less noise when compared to a standard DFB+EAM configuration.

Numerical results indicate that a gain up to 30 dB can be achieved using realistic parameters. This suggests that a custom-designed EML (DFB+EAM+DFB) could achieve an RF gain significantly higher than the measured value of 14 dB.

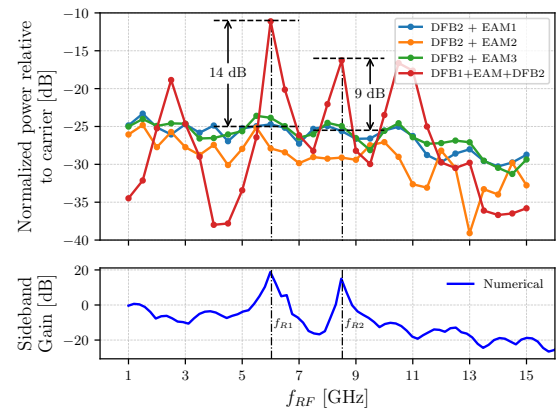


Fig. 4: Top: Measured sideband power as a function of the modulation frequency. Bottom: Numerical sideband gain for the experimental conditions.

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